

# Lucas A. Saavedra



Pueyrredón 1944  
Ramos Mejía, Buenos Aires, Argentina B1704  
+54 9 11 5574 2466  
lucasarielsaavedra@hotmail.com  
<https://lucassaavedra123.github.io/>

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## EDUCATION

**B.S. Software Engineering**, Pontificia Universidad Católica Argentina (UCA), Ciudad Autónoma de Buenos Aires, Argentina, March 2018-December 2023

## PROFESSIONAL EXPERIENCE

**Back-End Developer**, Deloitte, Ciudad Autónoma de Buenos Aires, Argentina, January 2021-June 2022

- Development of malware analysis tools.
- Python, Docker, MongoDB, and SQL were used for its development.

## ACADEMIC EXPERIENCE

**Research Assistant**, BIOMED Molecular Neurobiology Laboratory UCA-CONICET (director: Prof. Francisco J. Barrantes), Pontificia Universidad Católica Argentina (UCA), Ciudad Autónoma de Buenos Aires, Argentina, January 2022-Currently

- *The implementation of Deep Learning models is carried out for the analysis of the statics and dynamics of the nicotinic receptor of the neurotransmitter acetylcholine (nAChR) with orientation to graph theory, recurrent networks and geometric computing.*
- *The distribution and density of nicotinic acetylcholine receptors (nAChRs) in the plasma membrane of CHO-K1/A5 cells was studied using super-resolution microscopy (STED and STORM). Advanced analysis methods implemented with MATLAB based on graph theory and computational geometry were used to study the topography of nanoclusters.*
- *An efficient and autonomous algorithm based on binary classifications with supervised Graph Neural Networks (GNNs) was developed to detect and quantify the dynamic clustering of particles in membrane proteins in single-molecule microscopy (SMLM) data.*

## TEACHING EXPERIENCE

**Lecturer:** Data-oriented programming, Faculty of Engineering and Agricultural Sciences, Pontificia Universidad Católica Argentina (UCA), Ciudad Autónoma de Buenos Aires, Argentina, July 2024-Currently.

- *Oriented programming to objects .*
- *Technologies: Numpy , Pandas, Python, Java.*
- *Software design with UML.*

**Adjunct lecturer:** Discrete Mathematics, Faculty of Engineering and Agricultural Sciences, Pontificia Universidad Católica Argentina (UCA), Ciudad Autónoma de Buenos Aires, Argentina, March 2024-Currently.

- *Number theory.*
- *Graph theory.*
- *Demonstration methods.*
- *Boolean algebra.*

**Adjunct lecturer:** Introduction to data science, Faculty of Engineering and Agrarian Sciences, Pontificia Universidad Católica Argentina (UCA), Ciudad Autónoma de Buenos Aires, Argentina, March 2023-Currently.

- *Definitions and concepts of Artificial Intelligence.*
- *Introduction to artificial neural networks.*
- *Scientific method.*

## PUBLICATIONS / PRESENTATIONS / ABSTRACTS

Saavedra, L. A., Buena-Maizón, H., & Barrantes, F. J. (2022). Mapping the Nicotinic Acetylcholine Receptor Nanocluster Topography at the Cell Membrane with STED and STORM Nanoscopies. *International Journal of Molecular Sciences*, 23(18), 22.  
<https://doi.org/10.3390/ijms231810435>

Barrantes, F. J., Mosqueira, A., Buena-Maizón, H., & Saavedra, L. A. (2023). The individual molecule and its tribe, the nanocluster: the case of the nicotinic acetylcholine receptor. 11th International Weber Symposium.

Saavedra, L. A., Mosqueira, A., & Barrantes, F. J. (2024). A supervised graph-based deep learning algorithm to detect and quantify clustered particles. *Nanoscale*, 16(32), 15308-15318.  
<https://doi.org/10.1039/D4NR01944J>